Tanay Rathod

FPGA Design Engineer

RESUME_1205

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Total Experience: 10 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 10 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Intel Quartus, Timing Closure, DSP, AXI4, ILA, FPGA, VHDL, SPI, Python, Verilog, Xilinx Vivado, UART

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Lead FPGA Design Engineer | ZenithTrail Technologies | Jan 2023 - Present

- Verified 166 protocol corner cases with repeatable simulation suites
- Lowered end-to-end data-path latency by 24% using pipelined logic
- Reduced FPGA resource utilization by 21% through arithmetic sharing
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | KoraLink Innovations | Jan 2019 - Dec 2022

- Reduced FPGA resource utilization by 16% through arithmetic sharing
- Improved maximum operating frequency by 37% after timing-path optimization

Associate FPGA Design Engineer | Ironleaf Engineering | Jul 2016 - Dec 2018

- Verified 354 protocol corner cases with repeatable simulation suites
- Lowered end-to-end data-path latency by 23% using pipelined logic

PROJECTS

Real-Time Signal Filter | SPI, ILA, Timing Closure, DSP

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Protocol Adapter Core | FPGA, SPI, AXI4, Xilinx Vivado

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Meridian Technical University, 2016
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Digital Signal Processing Basics (2022)